

Focus Stacking with the Laplacian Pyramid

Burt & Adelson (1983) decomposition · energy-weighted band selection · one all-in-focus render

1 Capture the stack

Each frame is taken at a different focus distance. At macro magnification only a thin slab of the subject resolves.



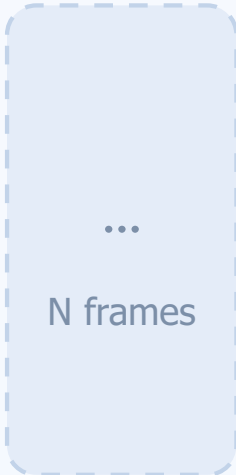
focus @ near plane



focus @ mid plane



focus @ far plane



Why one exposure is not enough

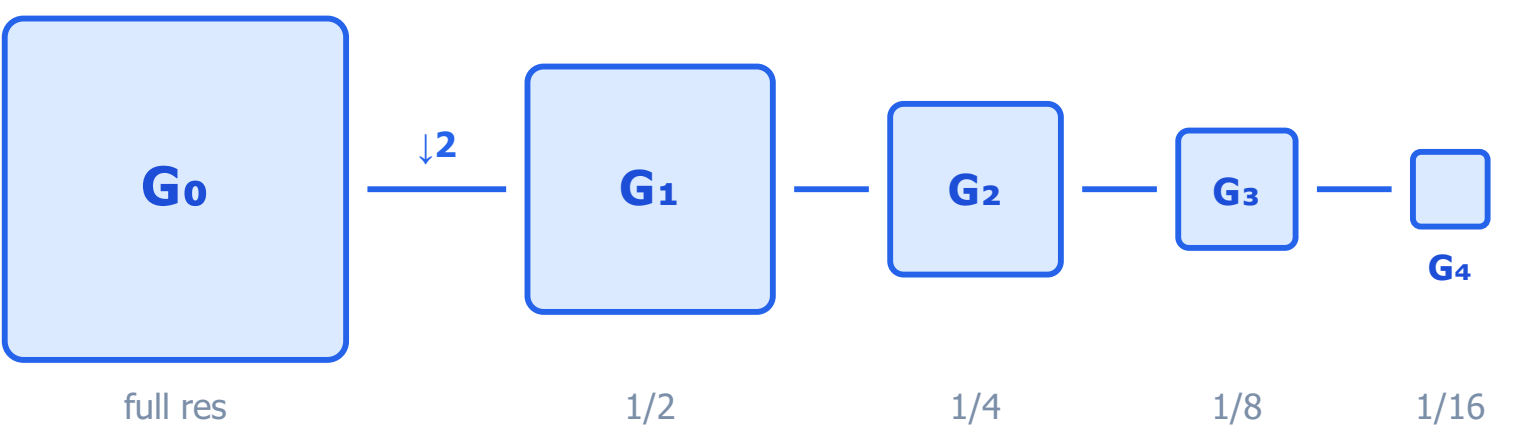
Depth of field shrinks with magnification. At 1:1 and f/8 it can fall below a millimetre, so no single aperture renders the whole subject — stopping down further trades focus for diffraction blur.

Solution: fuse many frames, not one.

2 Decompose — build the Laplacian pyramid

Every frame is split into scales: repeated 2×-downsampling gives a Gaussian pyramid; each band is what one level loses to the next.

GAUSSIAN PYRAMID — pyrDown, blur + decimate



LAPLACIAN BANDS — $L_i = G_i - \text{pyrUp}(G_{i+1})$



The pyramid is lossless — and that is the point

The bands add back exactly to the original image, so anything decided band by band can be reassembled without inventing pixels.

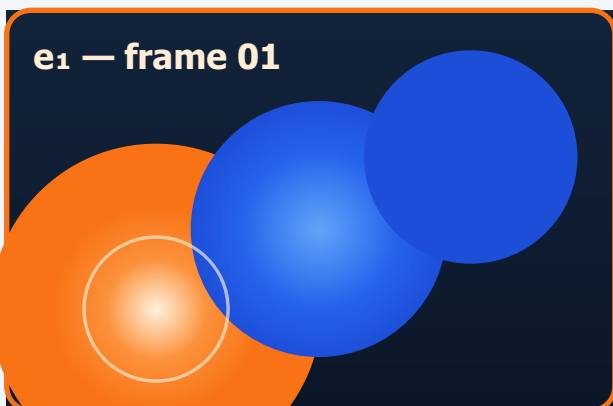
Each band answers a different question:

- L_0-L_1 — pore-level texture, grain, the finest hairs
- L_2-L_3 — contours, structural edges, coarse relief
- G_4 base — overall tone and colour, no detail at all

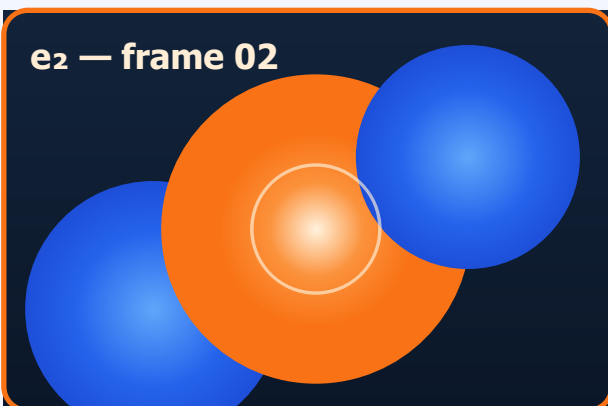
Working across scales is what stops a single sharp speck from dragging a whole region with it.

3 Measure focus — local energy per band

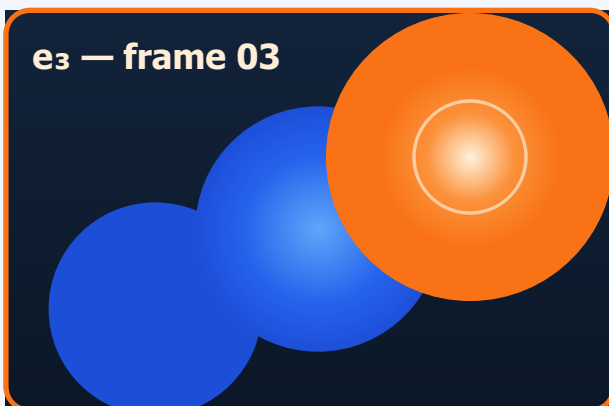
A single |Laplacian| pixel is noise. Pooling its square over a 5×5 window turns it into a region measure: a focus heatmap per frame, per band.



near plane wins here



mid plane wins here



far plane wins here

no detail resolved — defocus, flat tone, grain only

high band energy — genuinely in focus

Focus energy, normalised

$$e_i = \text{box}_{5 \times 5}(L_i^2) / n_i$$

Dividing by n_i , the frame's own noise level (10th percentile of its live band energies), stops a bright defocused veil from outscoring a dark sharp frame on grain alone. Every frame then reads about 1.0 where it resolves nothing — so those regions tie.

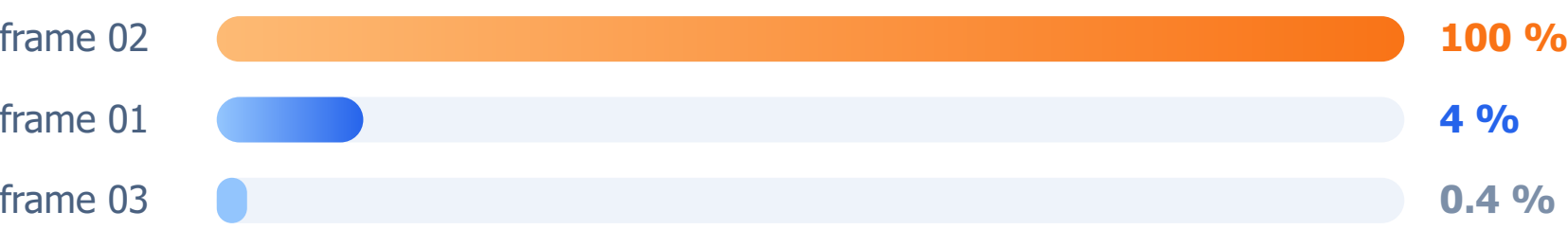
4 Select & merge — band by band, pixel by pixel

The classic rule copies the strongest coefficient. A soft exponent keeps that decision where it is real and averages where nothing is in focus.

Weighting rule

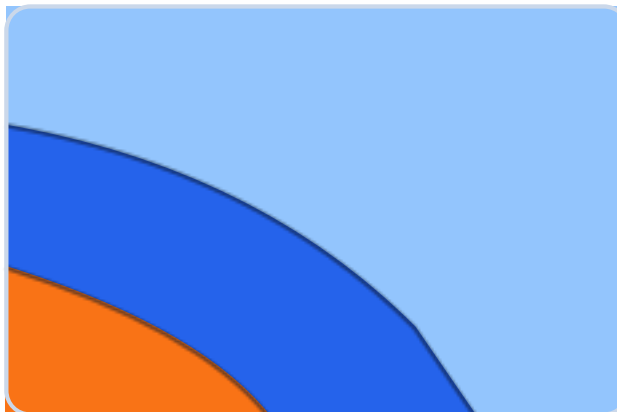
$$w_i = (e_i / e_{\max})^\beta \text{ with } \beta = 8$$

CONTRIBUTION TO THE FUSED BAND



A frame 2× behind the winner contributes $2^{-8} \approx 0.4\%$ — effectively a decision, without the speckle.

Selection map for one band



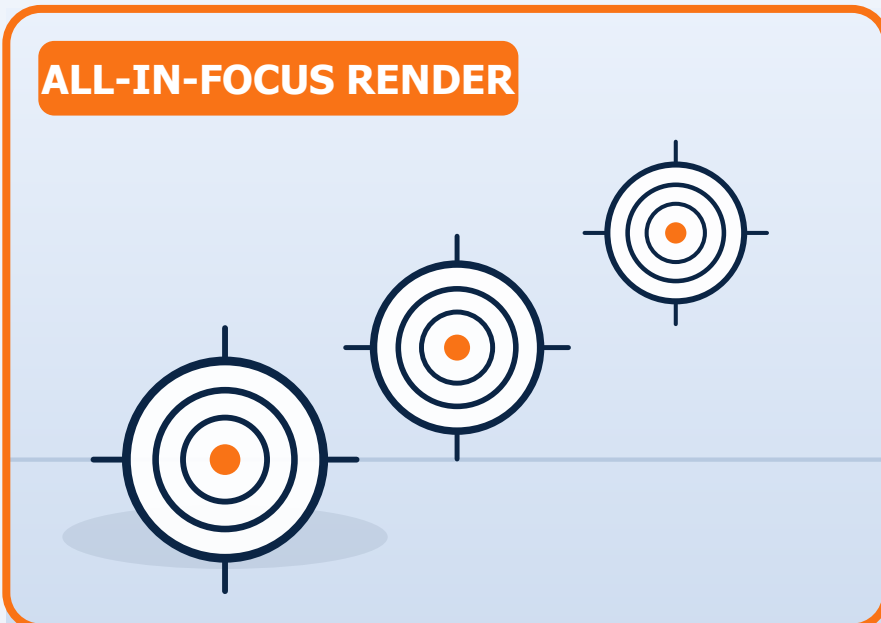
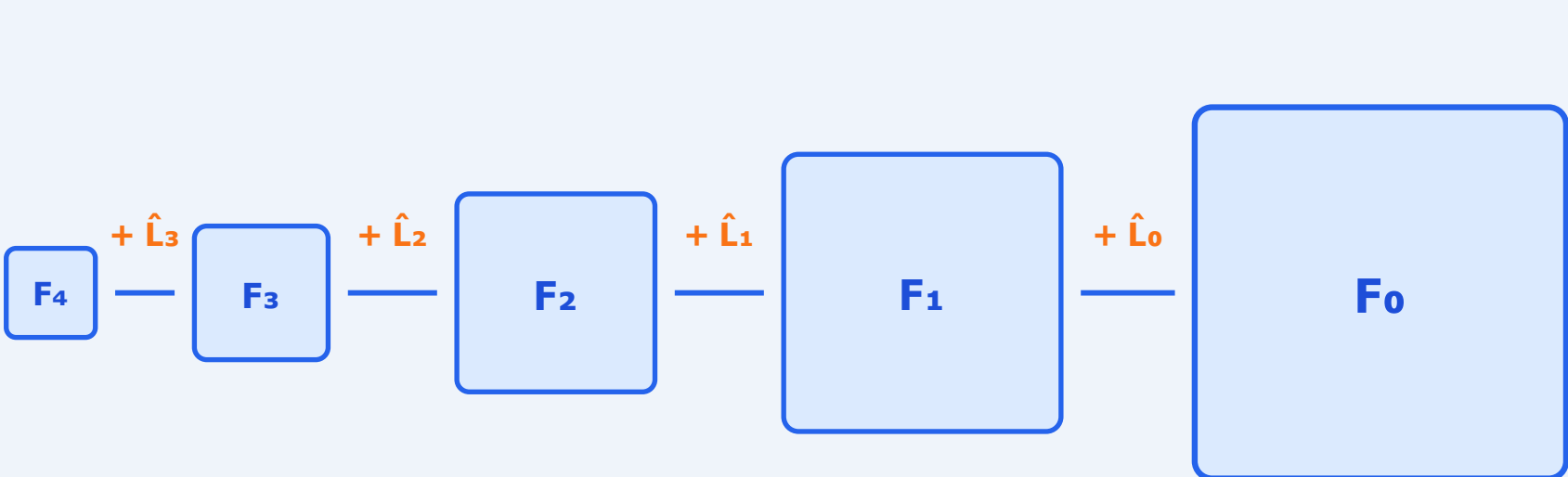
- frame 01 carries this region
- frame 02 carries this region
- frame 03 carries this region

Boundaries follow depth, not a grid.

Where frames tie, the weights average them — the correct answer for “nothing resolves here”.

5 Collapse the pyramid — one fully focused image

From the coarsest level down: upsample, add the fused band, repeat. The stack is gone; the detail of every frame is not.



PIPELINE AT A GLANCE

align → decompose → score
→ weight → merge → collapse

levels	5
pooling window	5 × 5
selectivity β	8.0

Front to back sharp — from a stack no single exposure could capture.